

Lecture 36 — Characteristic equation, 2×2 eigenproblems, and complex numbers

Wed Nov 18

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Reading: [Strang §6.1](#), characteristic equation and the complex-eigenvalue example. Related objectives: [Lectures](#).

Learning objectives

By the end of class, students should be able to:

1. Interpret $a + bi$ algebraically and as a point/vector in the complex plane.
2. Perform the basic complex arithmetic needed for eigenvalue calculations.
3. Form and solve $\det(A - \lambda I) = 0$ for a 2×2 matrix.
4. Find corresponding eigenvectors.
5. Check real or complex eigenpairs by substitution.